

Reproducibility Supplement

Innovation-Normalized Conformal Prediction for Autoregressive Episodes: Training and Calibration Costs

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1 Study scope and simulation protocol

This supplement specifies the simulation design, numerical integration settings, comparison methods and software environment used in the accompanying article. It also reports the complete comparison, allocation and sensitivity results. All experiments use synthetic episodes from the stated data-generating mechanisms.

The primary comparison settings were fixed in `reproducibility/PROTOCOL.md` before the reported runs. This is an internal protocol, not an external preregistration. A two-repetition-per-cell implementation pilot used seed 2026092802; pilot outcomes are excluded from the reported estimates. The primary study uses seed 2026092812. The optimizer sensitivity analysis is a subsequent analysis and is reported separately. Source code, random seeds, per-repetition outcomes and table-generation scripts are available in the accompanying repository.

2 Design and estimands

Each independent fitted repetition uses 500 training episodes, 499 calibration episodes and 1,000 evaluation episodes. The target miscoverage is .1, giving rank 450 and rank coverage .9. Histories have length 4 or 12. Four mechanisms and two history lengths, with 100 repetitions in each cell, give 800 independent fitted datasets. All eight procedures share the datasets within a cell. Separate random streams determine training, calibration, evaluation and model fitting.

The common marginal scale is $\sigma = \sqrt{3/G}$, $G \sim \text{Gamma}(4, 1)$, independent across episodes and independent of standardized shape. Thus $\mathbb{E}\sigma^2 = 1$ and $\mathbb{E}\sigma^4 = 1.5$. The standardized mechanisms are:

1. Gaussian stationary AR(1), coefficient .8 and innovation variance .36.
2. AR(1), coefficient .8, standardized Student- t_5 innovations multiplied by .6; discard 512 initialization steps. The resulting finite burn-in is an approximation to stationarity.
3. Gaussian AR(2), coefficients (.5,.3), innovation variance .4457142857142857; discard 512 initialization steps.
4. Gaussian AR(1), independent episode coefficient chosen equally from $\{-.8, .8\}$, with stationary initialization conditional on that coefficient.

Training, calibration and evaluation share the same standardized mechanism in each structural study. A paired amplitude arm doubles every coordinate in each evaluation episode. An additional Gaussian negative control doubles only the future response, leaving its history unchanged. This last arm does not satisfy the scale-transfer premise.

The primary estimands are the mean, over fitted datasets, of test coverage and mean interval length. The reported Monte Carlo standard errors use the 100 fitted repetitions as independent units. The 1,000 observations within a fitted test set are not treated as independent fitted models. For paired length means $\bar{X}, \bar{Y}$, let $r = \bar{X}/\bar{Y}$. The ratio standard error is

$$\widehat{\text{se}}(r) = \frac{s(X - rY)}{\sqrt{R}\bar{Y}}, \quad R = 100.$$

Intervals use $r \pm 1.96\widehat{\text{se}}(r)$ and are descriptive Monte Carlo intervals, not simultaneous inference across all comparisons. Width is zero for an empty set, and empty-set frequency is retained. Training times in the raw file are host-specific: AR ablations share the coefficient fit, and the Student timing records that fit only. These times are not used to rank methods.

3 Comparator specification and verification

IN-ARCP clips the pooled estimate to $[-.98, .98]$ and uses the history innovation scale. The raw-RMS ablation changes only that scale to $\sqrt{m^{-1} \sum_j H_j^2}$. Global AR calibration uses the same center and an unnormalized

absolute residual. The Student reference uses the fitted AR center and innovation scale with $t_{m,.95}$; it is an exact pivot only when the coefficient equals the true Gaussian parameter.

For invariant learned procedures, define $A(H) = \sqrt{m^{-1} \sum_j H_j^2}$, train on $(H/A(H), Y/A(H))$, and multiply predicted endpoints by $A(H)$ on the original scale. Native Ad-EffOort instead receives the original coordinates. Both use an intercept and all history coordinates. The independently written smoothed absolute-error-quantile gradient uses 1,000 updates, steps $t^{-.6}$, smoothing .1, and compact polynomial weights proportional to $[1 - (d/.1)^2]^2$ for $|d| < .1$. Its residual quantile forest has 100 trees, maximum depth 5 and quantile .9. This follows the published four-step construction with stated settings; it does not assert optimal tuning or exact reproduction of the authors' benchmark.

The vectorized linear update was compared with the authors' `utils.py` at commit 025118446ab9636837e91602efe4fa5e138f2373 (Git blob 58d4bf1c69002d0eab982fa5b25beb574434bf71). In a three-seed numerical comparison, initial updates agreed within 2.3×10^{-16} , while the final coefficient difference after 1,000 updates reached .229 on one dataset. Algebraically equivalent polynomial evaluations and different reduction orders can therefore lead to different finite-step trajectories in this nonsmooth optimization. The experiment compares the specified implementation of the published construction, rather than an exact replication of the authors' benchmark.

A subsequent sensitivity analysis uses the same 200 Gaussian fitted datasets as the primary study. It evaluates 1,000 and 2,000 updates with both simplified polynomial weights and the unscaled polynomial expression used upstream. Both versions use vectorized reductions. All 16 aggregated combinations are reported, including the primary simplified/1,000-step configuration. No replacement configuration is selected using evaluation performance. The upstream illustrative notebook uses some different settings, including a .8 residual quantile; the present study uses .9 consistently with its target coverage.

Invariant CQR uses a quantile forest with 100 trees, depth 5, quantiles .05 and .95, followed by corrected additive calibration of $\max(\hat{q}_{.05}(H) - Y, Y - \hat{q}_{.95}(H))$. The invariant nonlinear RF procedure uses a center forest with 100 trees, depth 8 and minimum leaf size 10; a residual-scale forest has 100 trees, depth 5 and minimum leaf size 10. Its absolute normalized residual is calibrated multiplicatively. No hyperparameter is selected on the evaluation episodes. These comparisons address specified implementations and information sets, rather than the best achievable performance of every method family.

4 Numerical mathematics and planning

Whitening, determinant and directional-variance identities are checked over $m \in \{2, 3, 4, 12, 25\}$ and $\rho \in \{-.95, -.4, 0, .4, .95\}$. Known-coefficient calibration means are checked over $m \in \{2, 4, 12\}$ and calibration sizes

$$n \in \{9, 19, 49, 50, 99, 100, 199, 200, 499, 1999\}.$$

The Jacobi rules have 128 and 512 nodes and are compared with independently implemented adaptive beta-density integration. The public utility defaults to adaptive tail integration because inverse-CDF quadrature converges slowly at extreme ranks. No fixed node count is claimed universally accurate.

Four complete fitted-mean calculations use $m = 2$. The coarse integration uses 64 coefficient bins, 256 midpoint angles and 32 beta nodes; the refinement uses 128 bins, 512 angles and 64 beta nodes. Both clipped coefficient atoms are retained. Each mean is checked against 30,000 independent fits and 32 test observations per fit, with episodes generated using a covariance Cholesky factor. Seed: 2026092801. The maximum discrepancy is 1.41 Monte Carlo standard errors. Grid refinement is a numerical diagnostic, not a certified error bound.

The local population curvature is checked at five coefficients with 4,096 circle directions and coefficient differences 10^{-4} . Coefficient MSE is evaluated using 10,000 repetitions at every (m, κ, N) in $\{4, 12\} \times \{1, 1.5, 3\} \times \{20, 100, 500\}$. Results include the strongly heterogeneous $\kappa = 3$ setting.

The allocation experiment uses a total of 300 episodes, $m = 4$, $\rho = .8$ and $\kappa \in \{1, 1.5\}$. Planning parameters are fixed from the synthetic design. There are 1,000 paired fitted repetitions per candidate, with 500 evaluation episodes, seed stream based on 2026092803. Candidate training sizes are 21, 51, 71, 101, 150, 201, 251, plus the planning recommendation 61 when $\kappa = 1$. A shared 300-episode pool is partitioned at each candidate boundary. The observed test outcomes do not choose the reported planning candidate. The full candidate table is supplied below; it does not establish a unique finite-sample optimum.

5 Software and execution

The saved comparison was run with Python 3.12.14, NumPy 2.3.5, SciPy 1.17.0, scikit-learn 1.8.0 and sklearn-quantile 0.1.1. Figure generation used Matplotlib 3.10.8. The four-worker comparison took approximately 848 seconds on the execution host; runtime is not a hardware-independent claim. Exact floating-point and model-fitting outcomes may depend on platform and package builds.

From the repository root, install `reproducibility/requirements.txt` and the package, then run the following commands. Existing saved outputs support rebuilding figures and PDFs without rerunning the simulations.

```
python -m pip install -r reproducibility/requirements.txt
python -m pip install -e .
python -m unittest discover -s tests -v
python reproducibility/validate_mathematics.py
python reproducibility/validate_expansion.py
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 \
  python reproducibility/run_comparisons.py --repetitions 100 --workers 4
python reproducibility/run_planning.py
python reproducibility/build_artifacts.py
python reproducibility/check_optimizer_sensitivity.py
python reproducibility/build_optimizer_summary.py
python reproducibility/build_supplement.py
cd paper
latexmk -pdf -interaction=nonstopmode -halt-on-error manuscript.tex
latexmk -pdf -interaction=nonstopmode -halt-on-error supplement.tex
```

The main manuscript uses the IEEE Access document class and IEEE numbered references. This supplement uses a separate, single-column layout for the full result tables. Template dependencies and their provenance are documented in `paper/TEMPLATE.md`. SHA256SUMS records the distributed evidence files. The numerical validation scripts evaluate the analytical identities and the consistency of the algorithm descriptions with the implementation.

6 Algorithm implementation and validation

Algorithms 1 and 2 of the article specify the prediction procedure and integer allocation search. Figure 1 represents the Python call sequence. The lifelines represent a caller, an `INARCP` instance and the module containing `history_scale`.

The executable check `reproducibility/validate_algorithm_description.py` compares the direct pseudocode formulas with the public implementation, including finite ranks, an unattainable rank, clipping, and the planning search. It also records normalization calls to verify their order, checks rejection of a zero test history even with an infinite threshold, and checks calibration invalidation after refitting. These tests assess implementation consistency; the sampling assumptions are specified by the experimental design.

7 Complete comparison results

“same” is the unchanged evaluation amplitude, “scale_x2” doubles the whole episode, and “future_x2” doubles only its response. Parentheses contain Monte Carlo standard errors. The ratio numerator is always IN-ARCP. A displayed zero may represent a positive empty-set frequency below the rounding precision. The machine-readable files preserve all digits and all per-fit measurements.

Method	Coverage (MCSE)	Length (MCSE)	IN/method [95% MC]	Empty
gaussian, $m = 4$, same				
IN-ARCP	0.8985 (0.0018)	2.318 (0.014)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.8998 (0.0019)	2.799 (0.020)	0.828 [0.819, 0.837]	0.000
AR+global	0.8978 (0.0018)	1.933 (0.009)	1.199 [1.184, 1.213]	0.000
Student	0.8988 (0.0010)	2.309 (0.004)	1.004 [0.992, 1.016]	0.000
Ad-EffOrt native	0.8975 (0.0020)	2.017 (0.010)	1.149 [1.136, 1.162]	0.000
Ad-EffOrt invariant	0.9002 (0.0017)	2.774 (0.018)	0.835 [0.825, 0.845]	0.000
CQR invariant	0.8977 (0.0019)	2.825 (0.021)	0.820 [0.811, 0.830]	0.000
RF invariant	0.8990 (0.0018)	2.649 (0.018)	0.875 [0.865, 0.885]	0.000
gaussian, $m = 4$, scale_x2				
IN-ARCP	0.8985 (0.0018)	4.635 (0.028)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.8998 (0.0019)	5.599 (0.041)	0.828 [0.819, 0.837]	0.000
AR+global	0.6203 (0.0026)	1.933 (0.009)	2.398 [2.369, 2.427]	0.000
Student	0.8988 (0.0010)	4.619 (0.008)	1.004 [0.992, 1.016]	0.000
Ad-EffOrt native	0.7016 (0.0035)	2.490 (0.018)	1.862 [1.832, 1.892]	0.000
Ad-EffOrt invariant	0.9002 (0.0017)	5.548 (0.036)	0.835 [0.825, 0.845]	0.000
CQR invariant	0.8977 (0.0019)	5.650 (0.041)	0.820 [0.811, 0.830]	0.000
RF invariant	0.8990 (0.0018)	5.298 (0.036)	0.875 [0.865, 0.885]	0.000

Method	Coverage (MCSE)	Length (MCSE)	IN/method [95% MC]	Empty
gaussian, $m = 4$, future_x2				
IN-ARCP	0.5708 (0.0031)	2.318 (0.014)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.6359 (0.0036)	2.799 (0.020)	0.828 [0.819, 0.837]	0.000
AR+global	0.5377 (0.0025)	1.933 (0.009)	1.199 [1.184, 1.213]	0.000
Student	0.5700 (0.0016)	2.309 (0.004)	1.004 [0.992, 1.016]	0.000
Ad-EffOort native	0.5424 (0.0029)	2.017 (0.010)	1.149 [1.136, 1.162]	0.000
Ad-EffOort invariant	0.6309 (0.0035)	2.774 (0.018)	0.835 [0.825, 0.845]	0.000
CQR invariant	0.6292 (0.0036)	2.825 (0.021)	0.820 [0.811, 0.830]	0.000
RF invariant	0.6148 (0.0034)	2.649 (0.018)	0.875 [0.865, 0.885]	0.000
gaussian, $m = 12$, same				
IN-ARCP	0.8991 (0.0015)	2.008 (0.008)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.8986 (0.0014)	2.253 (0.010)	0.891 [0.884, 0.898]	0.000
AR+global	0.8972 (0.0017)	1.923 (0.009)	1.044 [1.036, 1.052]	0.000
Student	0.9006 (0.0010)	2.012 (0.002)	0.998 [0.991, 1.005]	0.000
Ad-EffOort native	0.8975 (0.0018)	2.044 (0.009)	0.983 [0.974, 0.991]	0.000
Ad-EffOort invariant	0.9001 (0.0015)	2.387 (0.012)	0.841 [0.832, 0.850]	0.000
CQR invariant	0.8985 (0.0015)	2.358 (0.010)	0.852 [0.845, 0.859]	0.000
RF invariant	0.9001 (0.0016)	2.318 (0.011)	0.866 [0.858, 0.874]	0.000
gaussian, $m = 12$, scale_x2				
IN-ARCP	0.8991 (0.0015)	4.016 (0.016)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.8986 (0.0014)	4.507 (0.020)	0.891 [0.884, 0.898]	0.000
AR+global	0.6175 (0.0026)	1.923 (0.009)	2.088 [2.071, 2.105]	0.000
Student	0.9006 (0.0010)	4.024 (0.004)	0.998 [0.991, 1.005]	0.000
Ad-EffOort native	0.7470 (0.0029)	2.868 (0.021)	1.400 [1.379, 1.422]	0.000
Ad-EffOort invariant	0.9001 (0.0015)	4.774 (0.024)	0.841 [0.832, 0.850]	0.000
CQR invariant	0.8985 (0.0015)	4.715 (0.021)	0.852 [0.845, 0.859]	0.000
RF invariant	0.9001 (0.0016)	4.635 (0.021)	0.866 [0.858, 0.874]	0.000
gaussian, $m = 12$, future_x2				
IN-ARCP	0.5223 (0.0022)	2.008 (0.008)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.5597 (0.0022)	2.253 (0.010)	0.891 [0.884, 0.898]	0.000
AR+global	0.5350 (0.0024)	1.923 (0.009)	1.044 [1.036, 1.052]	0.000
Student	0.5235 (0.0015)	2.012 (0.002)	0.998 [0.991, 1.005]	0.000
Ad-EffOort native	0.5449 (0.0025)	2.044 (0.009)	0.983 [0.974, 0.991]	0.000
Ad-EffOort invariant	0.5788 (0.0028)	2.387 (0.012)	0.841 [0.832, 0.850]	0.000
CQR invariant	0.5725 (0.0025)	2.358 (0.010)	0.852 [0.845, 0.859]	0.000
RF invariant	0.5702 (0.0024)	2.318 (0.011)	0.866 [0.858, 0.874]	0.000
heavy, $m = 4$, same				
IN-ARCP	0.8991 (0.0017)	2.358 (0.016)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.9022 (0.0016)	2.786 (0.020)	0.847 [0.836, 0.857]	0.000
AR+global	0.8988 (0.0018)	1.846 (0.011)	1.278 [1.258, 1.297]	0.000
Student	0.8882 (0.0010)	2.217 (0.004)	1.064 [1.050, 1.078]	0.000
Ad-EffOort native	0.8983 (0.0016)	1.959 (0.012)	1.204 [1.187, 1.221]	0.000
Ad-EffOort invariant	0.8996 (0.0018)	2.835 (0.025)	0.832 [0.820, 0.843]	0.000
CQR invariant	0.9004 (0.0016)	2.905 (0.020)	0.812 [0.802, 0.822]	0.000
RF invariant	0.8995 (0.0016)	2.713 (0.022)	0.869 [0.858, 0.880]	0.000
heavy, $m = 4$, scale_x2				
IN-ARCP	0.8991 (0.0017)	4.717 (0.033)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.9022 (0.0016)	5.572 (0.040)	0.847 [0.836, 0.857]	0.000
AR+global	0.6642 (0.0031)	1.846 (0.011)	2.555 [2.517, 2.594]	0.000
Student	0.8882 (0.0010)	4.435 (0.009)	1.064 [1.050, 1.078]	0.000
Ad-EffOort native	0.7303 (0.0029)	2.438 (0.021)	1.935 [1.900, 1.970]	0.000
Ad-EffOort invariant	0.8996 (0.0018)	5.670 (0.051)	0.832 [0.820, 0.843]	0.000
CQR invariant	0.9004 (0.0016)	5.810 (0.041)	0.812 [0.802, 0.822]	0.000
RF invariant	0.8995 (0.0016)	5.427 (0.044)	0.869 [0.858, 0.880]	0.000

Method	Coverage (MCSE)	Length (MCSE)	IN/method [95% MC]	Empty
heavy, $m = 12$, same				
IN-ARCP	0.9029 (0.0017)	2.017 (0.012)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.9018 (0.0016)	2.235 (0.014)	0.903 [0.895, 0.910]	0.000
AR+global	0.8988 (0.0018)	1.829 (0.011)	1.103 [1.090, 1.116]	0.000
Student	0.8978 (0.0010)	1.962 (0.003)	1.028 [1.016, 1.040]	0.000
Ad-EffOrr native	0.9001 (0.0016)	1.996 (0.012)	1.011 [0.999, 1.023]	0.000
Ad-EffOrr invariant	0.9006 (0.0015)	2.415 (0.014)	0.835 [0.827, 0.844]	0.000
CQR invariant	0.9020 (0.0016)	2.403 (0.015)	0.840 [0.832, 0.848]	0.000
RF invariant	0.9013 (0.0016)	2.312 (0.015)	0.873 [0.864, 0.881]	0.000
heavy, $m = 12$, scale_x2				
IN-ARCP	0.9029 (0.0017)	4.035 (0.025)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.9018 (0.0016)	4.470 (0.028)	0.903 [0.895, 0.910]	0.000
AR+global	0.6628 (0.0032)	1.829 (0.011)	2.206 [2.180, 2.232]	0.000
Student	0.8978 (0.0010)	3.925 (0.006)	1.028 [1.016, 1.040]	0.000
Ad-EffOrr native	0.7753 (0.0028)	2.853 (0.028)	1.414 [1.387, 1.442]	0.000
Ad-EffOrr invariant	0.9006 (0.0015)	4.830 (0.028)	0.835 [0.827, 0.844]	0.000
CQR invariant	0.9020 (0.0016)	4.805 (0.030)	0.840 [0.832, 0.848]	0.000
RF invariant	0.9013 (0.0016)	4.623 (0.029)	0.873 [0.864, 0.881]	0.000
ar2, $m = 4$, same				
IN-ARCP	0.8983 (0.0017)	2.666 (0.016)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.8980 (0.0017)	3.076 (0.021)	0.866 [0.857, 0.876]	0.000
AR+global	0.8996 (0.0018)	2.270 (0.011)	1.174 [1.160, 1.188]	0.000
Student	0.9014 (0.0009)	2.684 (0.005)	0.993 [0.983, 1.004]	0.000
Ad-EffOrr native	0.8991 (0.0015)	2.239 (0.009)	1.190 [1.177, 1.204]	0.000
Ad-EffOrr invariant	0.8989 (0.0017)	2.989 (0.020)	0.892 [0.882, 0.901]	0.000
CQR invariant	0.9007 (0.0018)	3.027 (0.019)	0.881 [0.872, 0.890]	0.000
RF invariant	0.8994 (0.0017)	2.877 (0.018)	0.927 [0.918, 0.935]	0.000
ar2, $m = 4$, scale_x2				
IN-ARCP	0.8983 (0.0017)	5.331 (0.032)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.8980 (0.0017)	6.153 (0.042)	0.866 [0.857, 0.876]	0.000
AR+global	0.6219 (0.0027)	2.270 (0.011)	2.348 [2.321, 2.376]	0.000
Student	0.9014 (0.0009)	5.368 (0.011)	0.993 [0.983, 1.004]	0.000
Ad-EffOrr native	0.7096 (0.0029)	2.818 (0.020)	1.892 [1.861, 1.923]	0.000
Ad-EffOrr invariant	0.8989 (0.0017)	5.979 (0.040)	0.892 [0.882, 0.901]	0.000
CQR invariant	0.9007 (0.0018)	6.053 (0.038)	0.881 [0.872, 0.890]	0.000
RF invariant	0.8994 (0.0017)	5.754 (0.037)	0.927 [0.918, 0.935]	0.000
ar2, $m = 12$, same				
IN-ARCP	0.9004 (0.0016)	2.364 (0.010)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.9019 (0.0016)	2.632 (0.013)	0.898 [0.891, 0.905]	0.000
AR+global	0.8980 (0.0015)	2.260 (0.009)	1.046 [1.037, 1.054]	0.000
Student	0.8981 (0.0010)	2.340 (0.003)	1.010 [1.002, 1.018]	0.000
Ad-EffOrr native	0.8990 (0.0014)	2.288 (0.010)	1.033 [1.024, 1.042]	0.000
Ad-EffOrr invariant	0.9029 (0.0018)	2.658 (0.016)	0.889 [0.880, 0.898]	0.000
CQR invariant	0.9001 (0.0017)	2.601 (0.014)	0.909 [0.900, 0.917]	0.000
RF invariant	0.9004 (0.0018)	2.536 (0.016)	0.932 [0.923, 0.941]	0.000
ar2, $m = 12$, scale_x2				
IN-ARCP	0.9004 (0.0016)	4.727 (0.021)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.9019 (0.0016)	5.263 (0.027)	0.898 [0.891, 0.905]	0.000
AR+global	0.6209 (0.0025)	2.260 (0.009)	2.091 [2.074, 2.109]	0.000
Student	0.8981 (0.0010)	4.681 (0.005)	1.010 [1.002, 1.018]	0.000
Ad-EffOrr native	0.7504 (0.0027)	3.207 (0.024)	1.474 [1.452, 1.496]	0.000
Ad-EffOrr invariant	0.9029 (0.0018)	5.316 (0.031)	0.889 [0.880, 0.898]	0.000
CQR invariant	0.9001 (0.0017)	5.202 (0.029)	0.909 [0.900, 0.917]	0.000
RF invariant	0.9004 (0.0018)	5.071 (0.033)	0.932 [0.923, 0.941]	0.000

Method	Coverage (MCSE)	Length (MCSE)	IN/method [95% MC]	Empty
variable, $m = 4$, same				
IN-ARCP	0.9008 (0.0016)	3.539 (0.019)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.9010 (0.0016)	3.546 (0.019)	0.998 [0.996, 1.000]	0.000
AR+global	0.9004 (0.0015)	3.250 (0.016)	1.089 [1.074, 1.104]	0.000
Student	0.9071 (0.0009)	3.619 (0.007)	0.978 [0.968, 0.987]	0.000
Ad-EffOrr native	0.8976 (0.0018)	2.518 (0.011)	1.406 [1.389, 1.422]	0.000
Ad-EffOrr invariant	0.9010 (0.0016)	3.352 (0.021)	1.056 [1.046, 1.066]	0.000
CQR invariant	0.9011 (0.0017)	3.230 (0.022)	1.096 [1.085, 1.107]	0.000
RF invariant	0.8999 (0.0015)	3.028 (0.020)	1.169 [1.156, 1.182]	0.000
variable, $m = 4$, scale_x2				
IN-ARCP	0.9008 (0.0016)	7.078 (0.038)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.9010 (0.0016)	7.092 (0.038)	0.998 [0.996, 1.000]	0.000
AR+global	0.6237 (0.0026)	3.250 (0.016)	2.178 [2.148, 2.207]	0.000
Student	0.9071 (0.0009)	7.238 (0.015)	0.978 [0.968, 0.987]	0.000
Ad-EffOrr native	0.7108 (0.0028)	3.248 (0.023)	2.179 [2.143, 2.215]	0.000
Ad-EffOrr invariant	0.9010 (0.0016)	6.703 (0.042)	1.056 [1.046, 1.066]	0.000
CQR invariant	0.9011 (0.0017)	6.459 (0.045)	1.096 [1.085, 1.107]	0.000
RF invariant	0.8999 (0.0015)	6.056 (0.040)	1.169 [1.156, 1.182]	0.000
variable, $m = 12$, same				
IN-ARCP	0.9020 (0.0017)	3.346 (0.015)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.9018 (0.0017)	3.349 (0.015)	0.999 [0.996, 1.001]	0.000
AR+global	0.8985 (0.0015)	3.217 (0.016)	1.040 [1.029, 1.050]	0.000
Student	0.8892 (0.0010)	3.211 (0.005)	1.042 [1.033, 1.050]	0.000
Ad-EffOrr native	0.8993 (0.0017)	2.596 (0.013)	1.289 [1.275, 1.302]	0.000
Ad-EffOrr invariant	0.9016 (0.0016)	3.000 (0.015)	1.115 [1.105, 1.126]	0.000
CQR invariant	0.9007 (0.0016)	2.694 (0.014)	1.242 [1.229, 1.254]	0.000
RF invariant	0.9019 (0.0016)	2.664 (0.013)	1.256 [1.242, 1.270]	0.000
variable, $m = 12$, scale_x2				
IN-ARCP	0.9020 (0.0017)	6.691 (0.030)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.9018 (0.0017)	6.699 (0.030)	0.999 [0.996, 1.001]	0.000
AR+global	0.6222 (0.0025)	3.217 (0.016)	2.080 [2.059, 2.101]	0.000
Student	0.8892 (0.0010)	6.422 (0.009)	1.042 [1.033, 1.050]	0.000
Ad-EffOrr native	0.7501 (0.0031)	3.659 (0.028)	1.829 [1.801, 1.856]	0.000
Ad-EffOrr invariant	0.9016 (0.0016)	6.000 (0.031)	1.115 [1.105, 1.126]	0.000
CQR invariant	0.9007 (0.0016)	5.389 (0.028)	1.242 [1.229, 1.254]	0.000
RF invariant	0.9019 (0.0016)	5.328 (0.026)	1.256 [1.242, 1.270]	0.000

8 Complete allocation results

A star marks the prespecified approximation-based recommendation. All rows use 1,000 repetitions. Coverage varies slightly with the integer conformal rank; 150/150 has rank coverage 136/151, while the rank-aligned candidates shown have .9. This discrete difference is retained in the planning objective.

κ	N	n	Coverage (MCSE)	Length (MCSE)	vs. equal [95% MC]
1.0	21	279	0.9008 (0.0007)	2.4540 (0.0063)	0.9957 [0.9908, 1.0006]
1.0	51	249	0.9009 (0.0007)	2.4465 (0.0066)	0.9927 [0.9881, 0.9972]
1.0	61*	239	0.9011 (0.0007)	2.4482 (0.0067)	0.9933 [0.9889, 0.9978]
1.0	71	229	0.9012 (0.0007)	2.4498 (0.0068)	0.9940 [0.9896, 0.9984]
1.0	101	199	0.9017 (0.0008)	2.4555 (0.0073)	0.9963 [0.9926, 1.0001]
1.0	150	150	0.9017 (0.0009)	2.4646 (0.0083)	1.0000 [1.0000, 1.0000]
1.0	201	99	0.9016 (0.0010)	2.4809 (0.0102)	1.0066 [1.0015, 1.0117]
1.0	251	49	0.9006 (0.0014)	2.5306 (0.0151)	1.0268 [1.0163, 1.0373]
1.5	21	279	0.8999 (0.0007)	2.3519 (0.0058)	1.0001 [0.9954, 1.0048]
1.5	51	249	0.8998 (0.0007)	2.3392 (0.0060)	0.9947 [0.9904, 0.9990]
1.5	71*	229	0.8999 (0.0007)	2.3385 (0.0061)	0.9944 [0.9903, 0.9985]
1.5	101	199	0.8997 (0.0008)	2.3364 (0.0066)	0.9935 [0.9899, 0.9971]
1.5	150	150	0.9004 (0.0009)	2.3517 (0.0078)	1.0000 [1.0000, 1.0000]
1.5	201	99	0.8988 (0.0011)	2.3545 (0.0099)	1.0012 [0.9960, 1.0064]
1.5	251	49	0.9006 (0.0014)	2.4242 (0.0143)	1.0308 [1.0207, 1.0409]

9 Coefficient MSE diagnostic

The limit is $(1 - \rho^2)\kappa/m$ at $\rho = .8$. Every row uses 10,000 fits. Convergence is not monotone or uniformly accurate at the evaluated finite sample sizes.

m	κ	N	$\widehat{NMSE}$	MCSE	Limit
4	1.0	20	0.09713	0.00156	0.09000
4	1.0	100	0.09319	0.00133	0.09000
4	1.0	500	0.09156	0.00132	0.09000
4	1.5	20	0.12750	0.00199	0.13500
4	1.5	100	0.13053	0.00197	0.13500
4	1.5	500	0.13225	0.00191	0.13500
4	3.0	20	0.16757	0.00284	0.27000
4	3.0	100	0.19780	0.00387	0.27000
4	3.0	500	0.21742	0.00363	0.27000
12	1.0	20	0.03210	0.00050	0.03000
12	1.0	100	0.03079	0.00045	0.03000
12	1.0	500	0.03084	0.00044	0.03000
12	1.5	20	0.04191	0.00064	0.04500
12	1.5	100	0.04468	0.00067	0.04500
12	1.5	500	0.04511	0.00064	0.04500
12	3.0	20	0.05513	0.00097	0.09000
12	3.0	100	0.06510	0.00111	0.09000
12	3.0	500	0.07558	0.00160	0.09000

10 Additional optimizer sensitivity analysis

The underlying Gaussian datasets, calibration sets and test episodes are identical to the primary comparison. “Invariant” is one for RMS-invariant preprocessing and zero for the native inputs. “Unscaled” denotes the upstream polynomial expression evaluated with vectorized reductions; it is stored as `canonical` in the CSV. Each cell has 100 repetitions. The main experiment’s simplified/1,000-step rows are retained alongside the three additional settings. These checks address arithmetic and iteration sensitivity for the specified linear construction, not optimal tuning or exact reproduction of the authors’ benchmark.

m	Invariant	Arithmetic	Steps	Coverage (MCSE)	Length	IN/method [95% MC]
4	0	unscaled	1000	0.8968 (0.0020)	2.010	1.153 [1.140, 1.166]
4	0	simplified	1000	0.8975 (0.0020)	2.017	1.149 [1.136, 1.162]
4	0	unscaled	2000	0.8967 (0.0020)	2.009	1.153 [1.140, 1.166]
4	0	simplified	2000	0.8974 (0.0020)	2.017	1.149 [1.136, 1.162]
4	1	unscaled	1000	0.8997 (0.0018)	2.773	0.836 [0.826, 0.846]
4	1	simplified	1000	0.9002 (0.0017)	2.774	0.835 [0.825, 0.845]
4	1	unscaled	2000	0.8996 (0.0018)	2.772	0.836 [0.826, 0.846]
4	1	simplified	2000	0.9001 (0.0017)	2.773	0.836 [0.826, 0.845]
12	0	unscaled	1000	0.8980 (0.0019)	2.049	0.980 [0.971, 0.989]
12	0	simplified	1000	0.8975 (0.0018)	2.044	0.983 [0.974, 0.991]
12	0	unscaled	2000	0.8986 (0.0018)	2.051	0.979 [0.970, 0.988]
12	0	simplified	2000	0.8978 (0.0018)	2.046	0.982 [0.973, 0.990]
12	1	unscaled	1000	0.8989 (0.0016)	2.386	0.842 [0.833, 0.851]
12	1	simplified	1000	0.9001 (0.0015)	2.387	0.841 [0.832, 0.850]
12	1	unscaled	2000	0.8988 (0.0016)	2.385	0.842 [0.833, 0.851]
12	1	simplified	2000	0.9000 (0.0015)	2.386	0.841 [0.833, 0.850]